

Functions Summary

An Introduction by Joshua Aaron

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1 Putting the Fun in Functions

The last few weeks we have been looking at ciphers, codes, and, more recently, figurative numbers. While these things are interesting in and of themselves, we were hoping to get some insight as to how the broad abstract mathematical idea of functions, well, functions in these contexts. Here we will take a step back and just go through all of the terms we have touched on and provide some narratives that might help us to grok these concepts. After every term we will include some examples and exercises. As always we highly recommend that you do the exercises, as there is really no other way to learn things other than by doing,¹ so exercises and “playing around” with definitions and trying to generate your own problems and questions is the best way to cement these ideas in your too-cool-for-school teenage skulls.² Without further ado, let’s get into it!

1.1 Sets and Fancy Symbols

Sets are one of the most basic building blocks of mathematics: they are essentially “bags” of items, which we call *elements*. These elements cannot be repeated and are often ordered (though not always. We say that a set is *finite* if we can count of each element at some point, and *countably infinite* if we can always talk about the idea of the “next” element, and *uncountably infinite* if we can’t even list them all!³

We often label sets with capitol letters like A or B and when we want to talk about all elements in a set we use a symbol $\forall$ to mean “for All,” and $\in$ to mean “in” so that $\forall a \in A$ is read “for all elements a in the set A .” When we want to say that an element of a specific type exists we use the $\exists$ symbol (read “there exists”). So if we want to say “there exists an element in the set, A which is purple” we might write that as $\exists a \in A$ s.t. a is purple.⁴ This might just look like annoying notation now, but it will be quite useful in the future (and saves us a lot of time writing “for all” and “there exists” and all that). Generally, understanding how to learn and even generate good notation is a powerful tool in your critical thinking arsenal: just as you probably can’t conceive of a world without decimals now that you have seen them for several years, good notation becomes part of the foundation of how one sees not only mathematics and technical fields, but the world at large.⁵

¹We have yet to figure out how to simply download information into our brains, but don’t worry, your teachers are working on it.

²do not attempt to put cement in your skull, this is unhealthy, and will most likely result in severely curtailed performance in math class.

³Most things we will be dealing with when it comes to functions are in fact infinite! This second type of infinity, uncountable infinity, is hard to wrap one’s mind around, we might tackle these ideas of ever larger infinities later in the year.

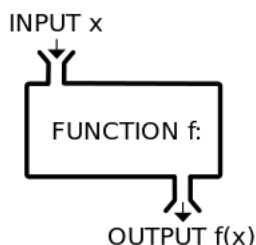
⁴note that s.t. is “such that” when we want to be really lazy

⁵It also helps us spot when, for example, jargon is used to try to confuse us—as is often the case in marketing or various financial situations: good notation is clear and helps us stay focused on what is important, bad notation is messy and takes a lot of brain power to process.

1.2 What is a Function?

Jumping From Set to Set We have actually been using and implicitly (and sometimes explicitly) talking about functions for the last month now, that’s good news right!? Nothing new to see here! Okay actually yes, there is as we were sometimes intentionally avoiding the use of standard notation and ways of thinking about functions so we had a chance to discover some of these things for ourselves. Here we will just go over some of the notation and names and conventions; some of which we have seen before, others of which we have consciously sidestepped so don’t worry if it looks new. Functions have at least three concrete ways of looking at them which we will discuss here: the key idea is that they are all sides of some funky functioney three-sided coin.⁶

Processes with Input and Output Functions can be thought of as *processes* that have a particular *input* or *argument* and an associated *output*. We might think of this as some kind of physical machine: like one where a kilo of chocolate gets poured in and an equally heavy chocolate Brandenburg Gate gets spit out after a few minutes⁷. We will look at several examples later, but as a quickie let’s just think about the (slightly boring) function that takes in a number and doubles it: we might think of this function as $f(x) = 2x$ that basically means for every quantity x you give me I will give you back twice as much.⁸ So $f(3) = 2 \times 3 = 6$ and $f(\frac{1}{2}) = \frac{1}{2} \times 2 = 1$. Almost like a recipe or a code we put something in and we consistently get a certain response.⁹ This we will call the process view of a function, basically that the function is a machine that takes something (usually a number—but sometimes a letter as we have seen before) as input and returns something (also usually a number) as output. We might visualize it like this:



But that is not the only way to look at functions! We can also think of them as a bunch of arrows from one set to another, where these arrows indicate a “relationship.”¹⁰

Life, Death, and Taxes: Relating Two Sets One of the reasons we study functions is that they are super useful when we want to relate the members of one set with another. A nice example of this is that everyone who works in Germany has a Tax Number: we want there to be just one number associated with each person, and we don’t want any two people to have the same number, how do we do that? Well, we could just count the number of people and then each person would have a number, but then maybe someone could guess my number based on birth records or some other fancy trickery, and then that could lead to trouble. So we want a “mapping” that makes it hard for someone to guess my number, but we want a consistent number length and form—not too long (something a human can remember) not too short (not something where I can guess all of them, or I run out of numbers in a couple years), and something that still works as people

⁶Which I am sure Schlomo has attempted to fabricate...

⁷If you have access to such a machine, please let us know!

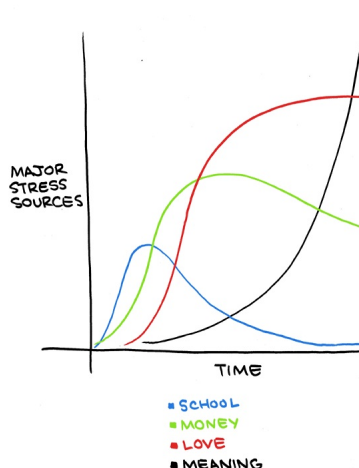
⁸This function immediately ceases to be boring when we combine it with our Chocolate Brandenburg Gate function—again, if you have a device which doubles the amount of chocolate it takes in as input, please let us know!

⁹Meaning every time I put in 3 into our $f(x)$, for example, I would always get 6, there (in the vast majority of cases) isn’t a random element to this—which makes sense if we think of functions as modeling computational processes, we don’t want our calculators giving us potentially random answers!

¹⁰Just not the lovey-dovey kind... those should be had with [people, not mathy things, unless the person is mathy, then that’s okay—actually probably a good idea as mathematicians tend to be happy people when it comes to job satisfaction!]

get born and die. This is a slightly different idea from the “process” or “number crunching machine” we just talked about—it is more about the relationship between the number and the person. This is where we see $f(x)$ as outputting something that we recognize as having a certain form: for example $f(x) = 2x$ should always give us even numbers, and putting in 2 gives us the “second” even number, 4.¹¹ Or we might think of just the super abstract case of a bunch of blobs with arrows pointing out of one and into another (we will come back to this later).

Pretty Pictures: Functions as Curves or Graphs One context in which we will often see functions is that of curve (or functional) analysis—this is where we are given some kind of graphs like these



and the key thing here is that we can describe these curves (which often have an *infinite* number of points in them!) just with a simple equation! You have probably seen some of these before in the form of linear equations (of the form $f(x) = mx + b$, for example $\text{double}(x) = 2x$) or quadratic equations (of the form $f(x) = ax^2 + bx + c$, for example $U(x) = (x + 2)^2 = x^2 + 4x + 4$). We will delve into these later on, the key idea is that we can talk about the behavior of these things from how they look both plotted with fancy axes but also just by analyzing the numbers and letters that make up their functional equation! That is the power of algebra!

So now that we have talked about this lets get into some key spaces that we will be dealing with when we talk about functions.

1.3 Funky Function Spaces:

Where We Start: New Domains A function’s *Domain* is the set from which we get our input—we can think of this like the birth place of our function—all the numbers start here and we end up somewhere else (sometimes) after sending them on the journey that is our function. For example, if we were talking about a function that takes a whole number and gives us an associated whole square number (which we might right $\text{square}(x) = x^2$ for $x \in \mathbb{Z}$ where $\mathbb{Z}$ are the integers.) We would say that this function has the integers as its *domain*. But we can square other numbers too right? So we might also choose to *restrict* or *broaden* the domain of this function, perhaps to include all real numbers ($\mathbb{R}$ —decimals including infinite decimals like π)! Or all rational numbers ($\mathbb{Q}$ —numbers that are fractions)! Sometimes, the domain is obvious, like in our cipher examples the domain was often our normal alphabet. But sometimes we have to think a little—once we have our domain we can also start talking about our *range*!

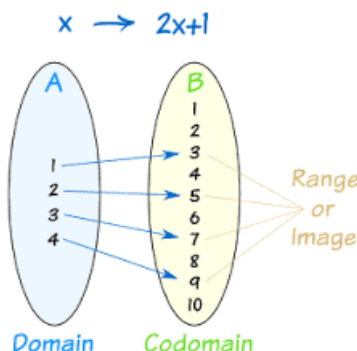
¹¹Or third if we are cool and start at 0, which is, incidentally, also even.

Just one more thing... Another word for *domain* is *pre-image*, which make the *range* the *image* of our function—this is how one talks about functions in German and it can be useful to think about a function as something which takes an initial picture and somehow distorts or changes it to something else ¹²

Home on the Range The *Range* of a function is the set of possible outputs the function can take—now this is sometimes tricky as it might not be the same as the domain (or set of possible inputs): if I have the function $double(x) = 2x$ which doubles whatever I put in, I might restrict my domain to the natural numbers, but then my *range* would be something different... can you see what it would be? If I only can put in $x \in \mathbb{N} = 1, 2, 3, \dots$ then I can't get any odd numbers out after putting them through $double(x)$! That means our *range* for $double(x)$ (when it's domain is restricted to the naturals) is *the set of even numbers*.

A Brief Aside So, you might read all that and say, “Why the hell would I care about that?” At which point, after pouring you a nice cup of tea to calm you down, I would simply say—but we do this all the time! Your bank card associates an address ¹³ with a person, you! And there is a rather limited set of numbers that are deemed acceptable credit card numbers, just as there are a limited number of addresses—so (assuming you only have one account per bank) here we have a function with domain of “possible existing addresses” and a range of “possible acceptable credit card numbers” and we want both of those to be restricted to meaningful manageable numbers and patterns that our soggy brains can handle!

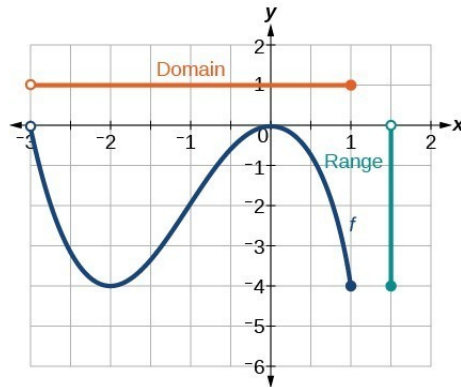
Back to the Range: The Lone Range...r So basically in the end, we have where our function starts (the domain) and what possible outputs our function has (the range). Some call the set that includes the range the *codomain* to point out that it might be that our range sits within a larger set (like how the evens sit within the natural numbers). Generally, worry less about the codomain, and more about the other terms, but in the end, we get a picture that looks like this:



When it comes to graphs that we plot (the snaky curvey things) the domain and range might look like this:

¹²In computer graphics, that is literally what one does to images using various often super interesting functions.

¹³Which, in the end, is just a number, yeah there are letters, but computers still encode them as numbers



What is next So now that we have talked a bit about what a functions are and some of the sensible spaces that we often refer to when we talk about them, let's briefly talk about what *is not* a function and why we by convention make this distinction. After that, we will offer up some exercises and then move on to discussing various different types of functions: this is a pattern we often have in math, we define a mathematical object, however roughly, and then we seek to understand what categories are useful within that new concept that we created. It is important to understand that most of this stuff is convention—meaning some of our notation and ideas are just accidents of history, but generally these ideas about functions survived *because they were useful!* We have seen when it comes to ciphers etc. but frankly, we ain't seen nuthin' yet—the applications of these things are quite literally to many to count!

1.4 What isn't a Function:

Definitions aren't very useful if everything falls into them¹⁴. So it is important to ask the question *what isn't a function?!?* Well, the smallest answer is: anything where you put in one (or more) inputs and get *more than one* output *is not* a function. Why the discrimination? Well, the answer is, we aren't really discriminating as there are indeed a whole class of objects that give multiple outputs, we normally call those *multi-functions* (looks like mathematicians weren't too creative here). But functions are particularly nice as they allow us to associate exactly one thing with one other thing—and this guarantee makes lots of calculations and analysis easier, it also models many things in the real world rather well—if I asked you to give me the longitude and latitude of where you were, you would give me exactly one point (with two coordinates—hold of on that for a sec), and if you gave me two points I would be concerned that you left part of yourself somewhere (which is usually unhealthy). Also, if our input is something that seems to only go in one direction like time¹⁵ then we might have our function give us something like “position” or “speed” or “number of falafel consumed” all of which usually have one possible value at a time¹⁶

Serious Vert So this leaves us with the following picture for what is and isn't a function: when we graph our function using schmancy coordinates, if we can draw a *vertical line* through our function we expect to only go through one point¹⁷ if it hits two points, then, unfortunately, it isn't a function by our definition. This leads to some interesting exclusions. Let's take two examples, the Square Root function ($\text{sqr}(x) = \sqrt{x}$) unfortunately gives us two outputs: for example $\text{sqr}(4) = 2$ and -2 ! But also circles are out—a circle centered at (a, b) with radius r can be written as $r^2 = (x + a)^2 + (y + b)^2$, but at the point $x = \frac{r}{2}$ we get two points as our answer! (check it out for yourself—maybe with a specific r and centered at the origin?) The

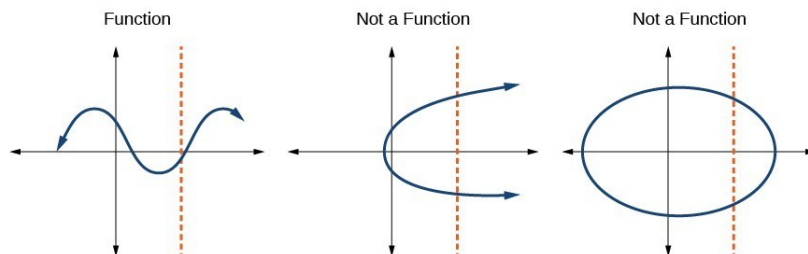
¹⁴What does that say about the definition of the word “everything?” Hmmm.... may have talked myself into a corner there...

¹⁵If you have found a way to go backwards in time or reverse it's direction, please let us know—there is much to discuss!

¹⁶If I have eaten exactly 3 falafel I have not also eaten exactly 8 falafel... though if you give me a few minutes I will get there, just ask Schlomo!

¹⁷Sometimes no points, but that's just sad :(I mean really... what's the point of drawing the line then?... get it... whats the point? Hehehe

picture looks like this for some sine like wave (a function!) a square root like curve, and a circle (both not functions):



Coordinating Chaos “But wait! There are lots of situations where we want to have two things come out as an answer! Like longitude and latitude, or maybe height and age of a basketball player, or the time and address of a concert!” Fantastic points all! Geez you are good—super nice and concrete examples—yes so we get around this restriction to *just one output* with things you may have heard of called coordinates¹⁸, vectors¹⁹, and matrices²⁰ So don’t worry, we aren’t really limiting ourselves here, we are just focusing on a smaller set of questions than the full existential depth of functions and what it means to be a function etc. With that in mind, let’s move on to describing various types of functions, hopefully you will see the relevance of these things to what we have been doing for the past weeks!

¹⁸Also called tuples which I just find to be a cute word no?

¹⁹glorified arrows

²⁰Numbers written in boxes that for some reason are super powerful and basically make the world run.

2 The -ives of Functions

Injective

Formal Definitions: every element in the domain has a specific unique partner in the range

Key Questions: Horizontal Line test

Examples

1. $bla(x) = x + 1$
2. $cube(x) = x^3$
3. $exp(x) = e^x$
4. Usernames (every User has a specific Username) not every username is taken
5. English to German alphabet (every english letter has a corresponding german letter, but not the other way around)

Non-Examples a function is not injective when two elements in the domain both map to the same element in the range (they both have the same image)

1. $square(x) = x^2$
2. $f(x) = \sqrt{x}$ define range!
3. $log(x)$
4. passwords (two people can have the same password)

Exercises

1. Steps to show injectivity
2. Steps to show a function *is not* injective
3. Polynomials
4. Function composition

Surjective

Formal Definitions: Every element of the range is associated with at least one element in the domain.

Key Questions: Image is fully covered by function

Examples

1. $cube(x) = x^3$
2. $bla(x) = x + 1$
3. $floor(x)$
4. Addresses (every house has a number, but sometimes we skip numbers)
- 5.

Non-Examples If we can't go backwards and get at least one option...

1. $\exp(x) = e^x$
2. $\text{square}(x) = x^2$
3. Algebraic fractions (with Poles/asymtotes)
4. $\sin(x)$,

Bijjective

Formal Definitions: there and back again, injective and surjective

Key Questions: does this function have a universal inverse function
are these sets the same size
Does this function map one set onto another

Examples

1. Even Numbers (or Odd Numbers) mapped to the naturals
2. $\text{bla}(x) = 2x + 3$ (all lines!)
3. $\text{cube}(x) = x^3$
4. Naturals and Reals (proof!)
5. our ciphers (shift ciphers)

Non-Examples

1. $\text{square}(x) = x^2$ Unless we restrict range
2. $\text{abs}(x) = |x|$ (again, unless we restrict the range
3. $\text{floor}(x)$ or $\text{ceiling}(x)$
4. Naturals and Reals (Proof?)

Some Other Key Examples

1. Good Ciphers are: Bijjective
2. Good Lossless Compression Codes are Bijjective
3. Good Lossy Compression Codes are Surjective
4. Good Error Correction Encoding Schemes Surjective
5. Good Error Correcting Decoding Schemes are injective
6. the figurate numbers (of a given n-gon) are all bijective to the natural numbers (if we make a function only between the set of natural numbers and the set of, for example, triangular numbers—otherwise it would only be injective as we see later)
7. every natural number has at least one associated figurate, so the function from figurates to numbers is surjective
8. The generating function for a given (n-gon) figurate number is injective (but obviously not surjective)